



CS 340: Algorithms and Data Structures

Introduction

Dr. Ali Moltajaei Farid
Dept. of Computer Science

CS 340

- Problem solving, with applications.
- Algorithm: method for solving a problem.
- Data structure: method to store information.

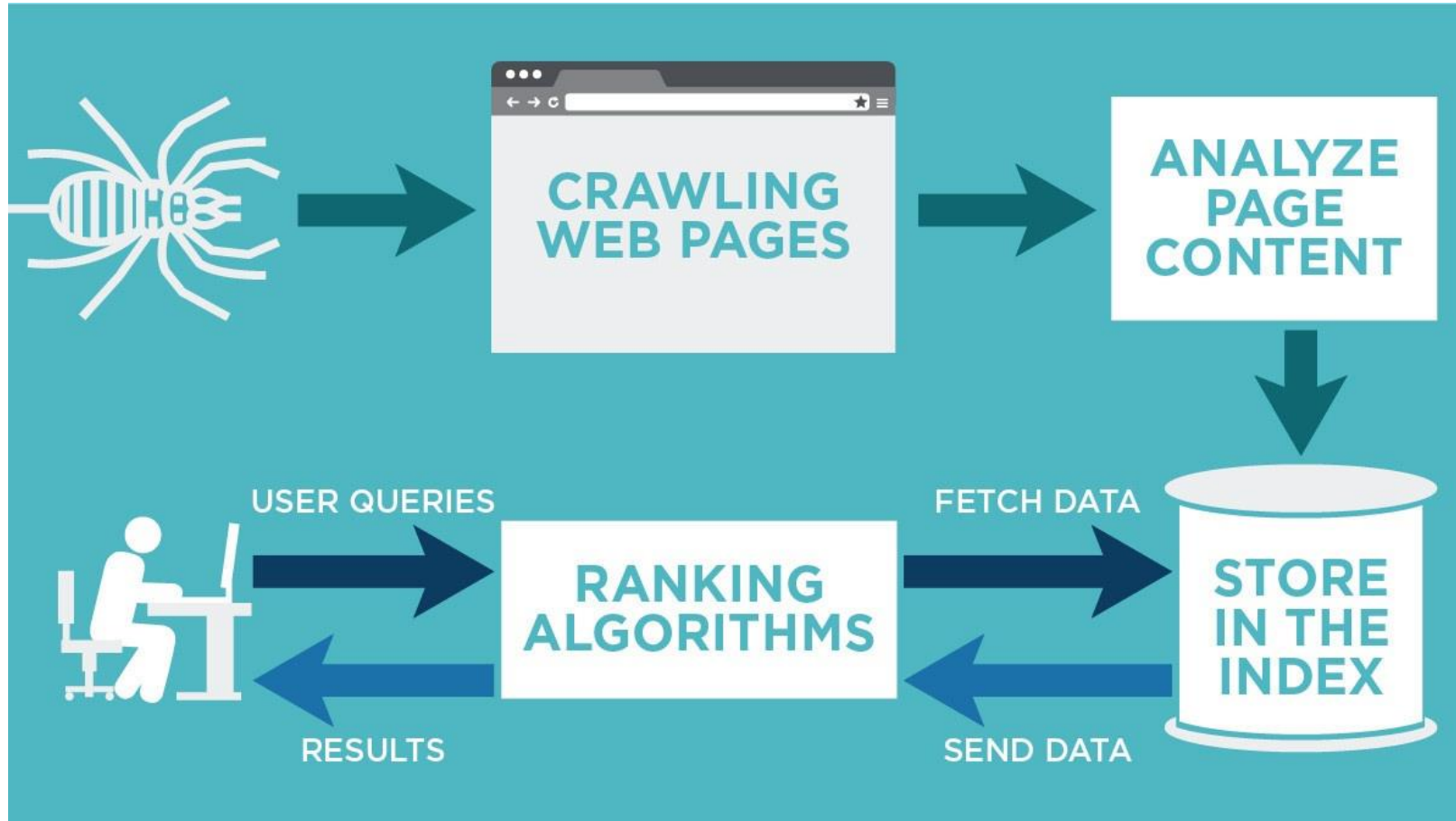
topic	data structures and algorithms
data types	stack, queue, bag, union-find, priority queue
sorting	quicksort, mergesort, heapsort, radix sorts
searching	BST, red-black BST, hash table
graphs	BFS, DFS, Prim, Kruskal, Dijkstra
strings	KMP, regular expressions, tries, data compression
advanced	B-tree, k-d tree, suffix array, maxflow

Why study algorithms?

Their impact is broad and far-reaching.

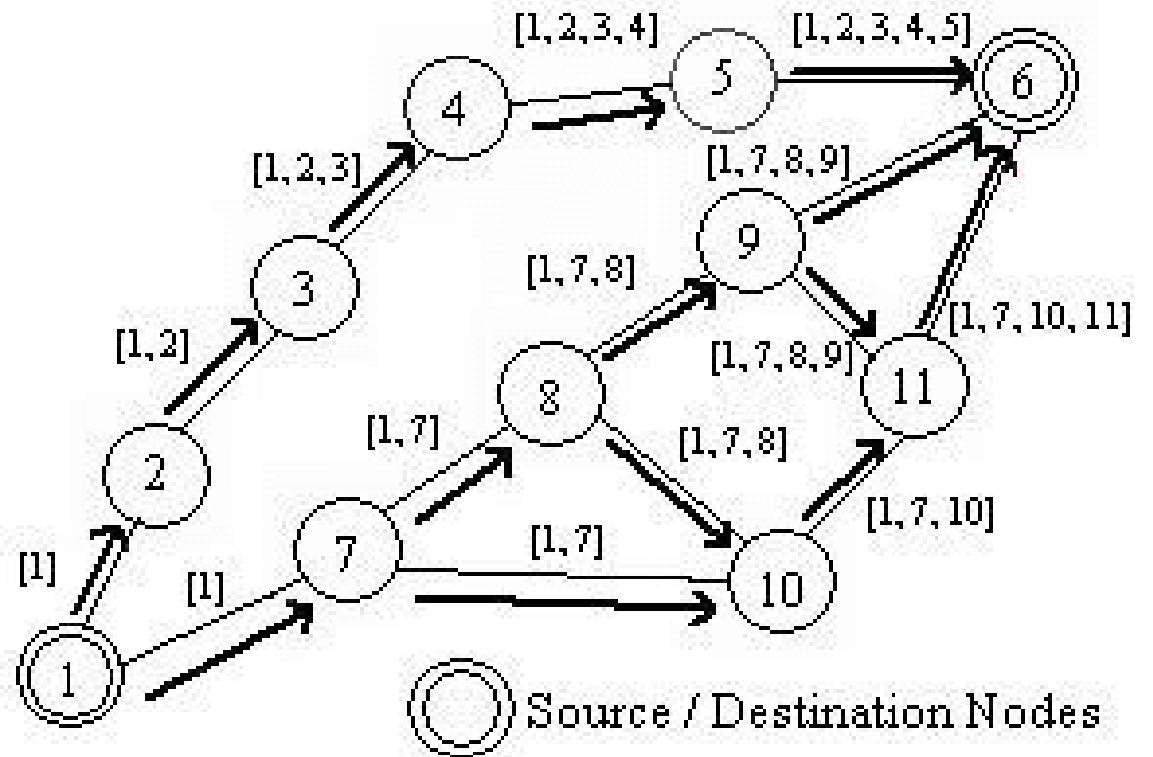
- Internet. Web search, packet routing, distributed file sharing,...
- Biology. Human genome project, protein folding, ...
- Computers. Circuit layout, file system, compilers, ...
- Computer graphics. Movies, video games, virtual reality, ...
- Security. Cell phones, e-commerce, voting machines, ...
- Multimedia. MP3, JPG, DivX, HDTV, face recognition, ...
- Social networks. Recommendations, news feeds, advertisements, ...
- Physics. N-body simulation, particle collision simulation, ...

Web search engine

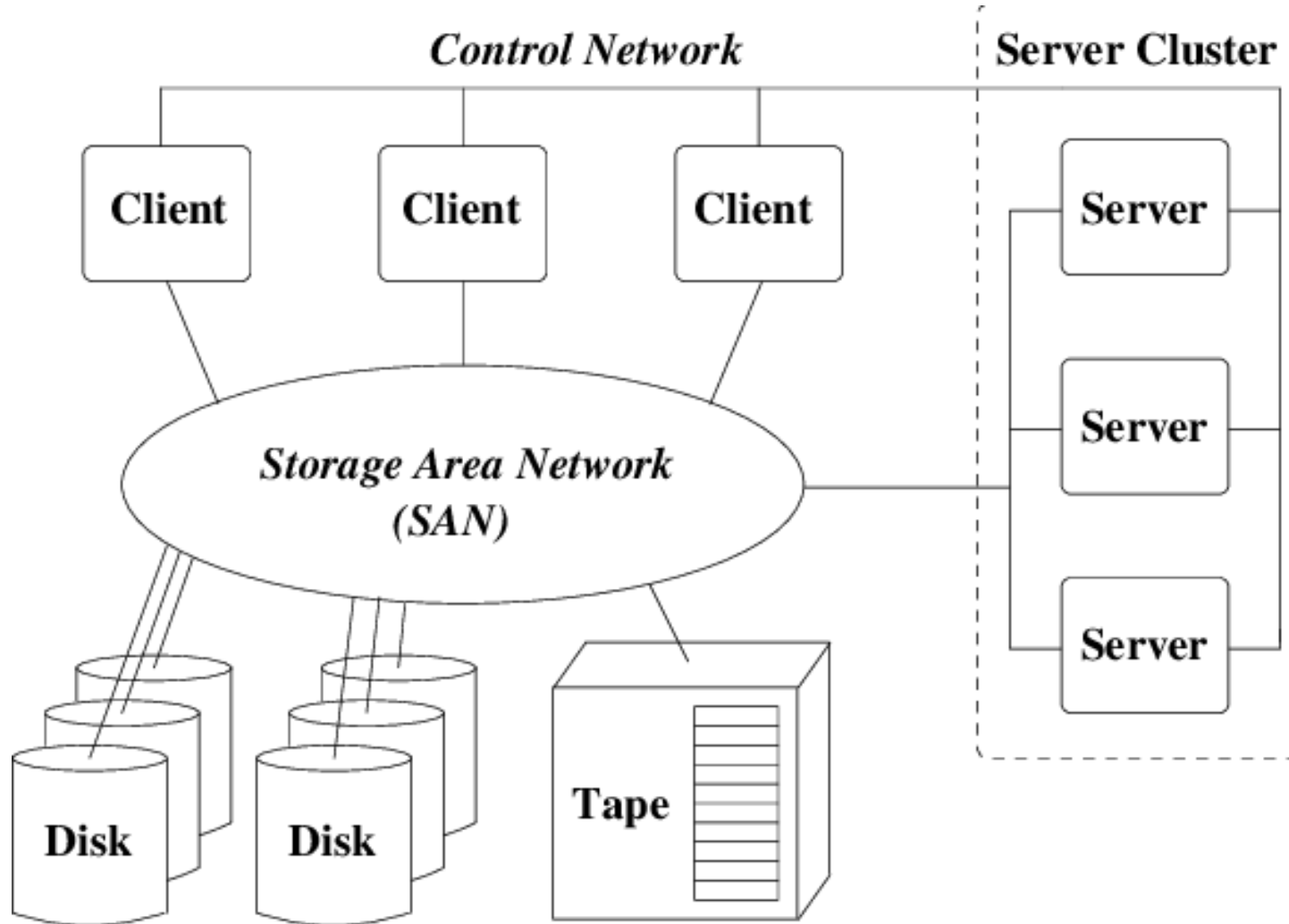


Packet routing

- The Internet enables people all around the world to quickly access and retrieve large amounts of information
- With the aid of clever algorithms, sites on the Internet are able to manage and manipulate this large volume of data.
- Examples of problems that make essential use of algorithms include finding good routes on which the data will travel

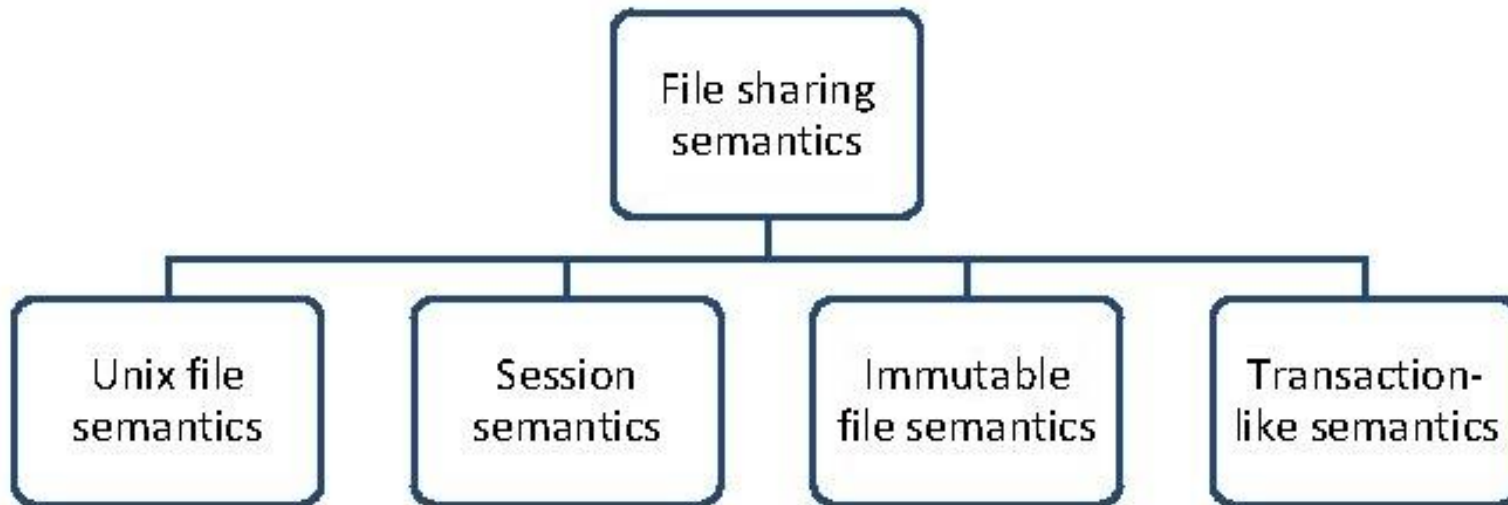


Distributed file sharing (1)



Distributed file sharing (2)

- A shared file may be simultaneously accessed by multiple users.
- In such a situation, an important design issue for any file system is to clearly define when modifications of file data made by a user are observable by other users.

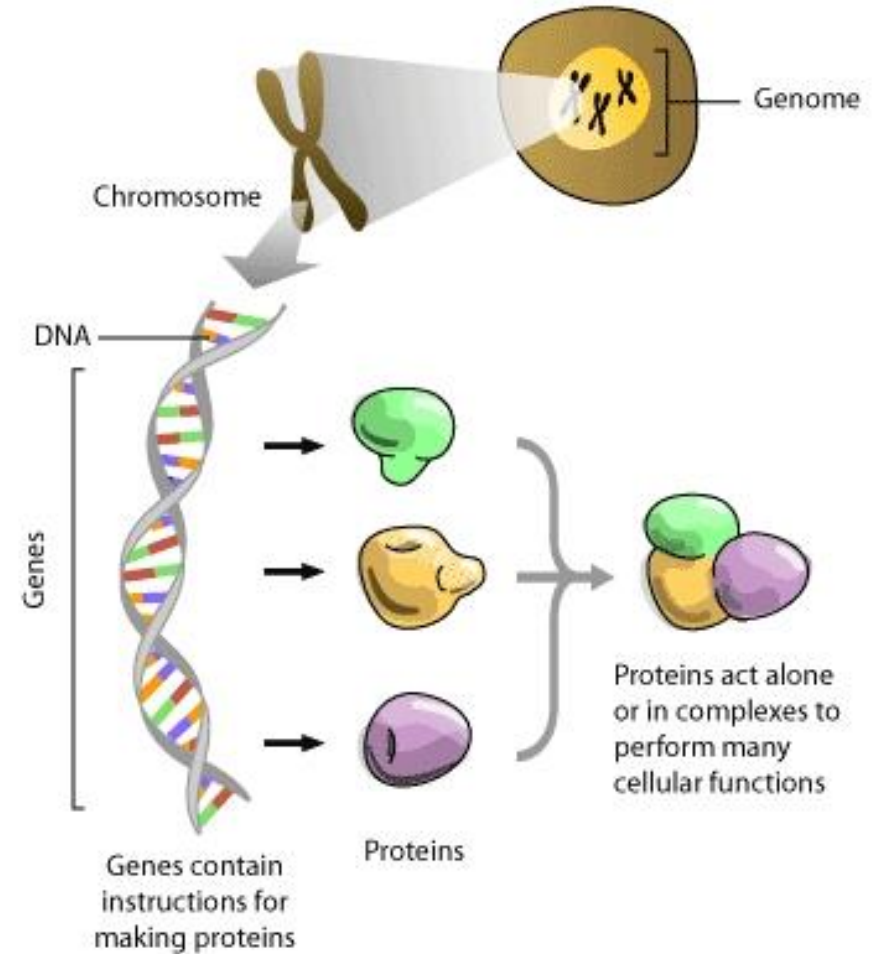
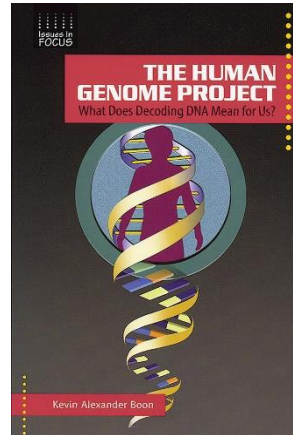


Human genome project

The Human Genome Project has made great progress toward the goals

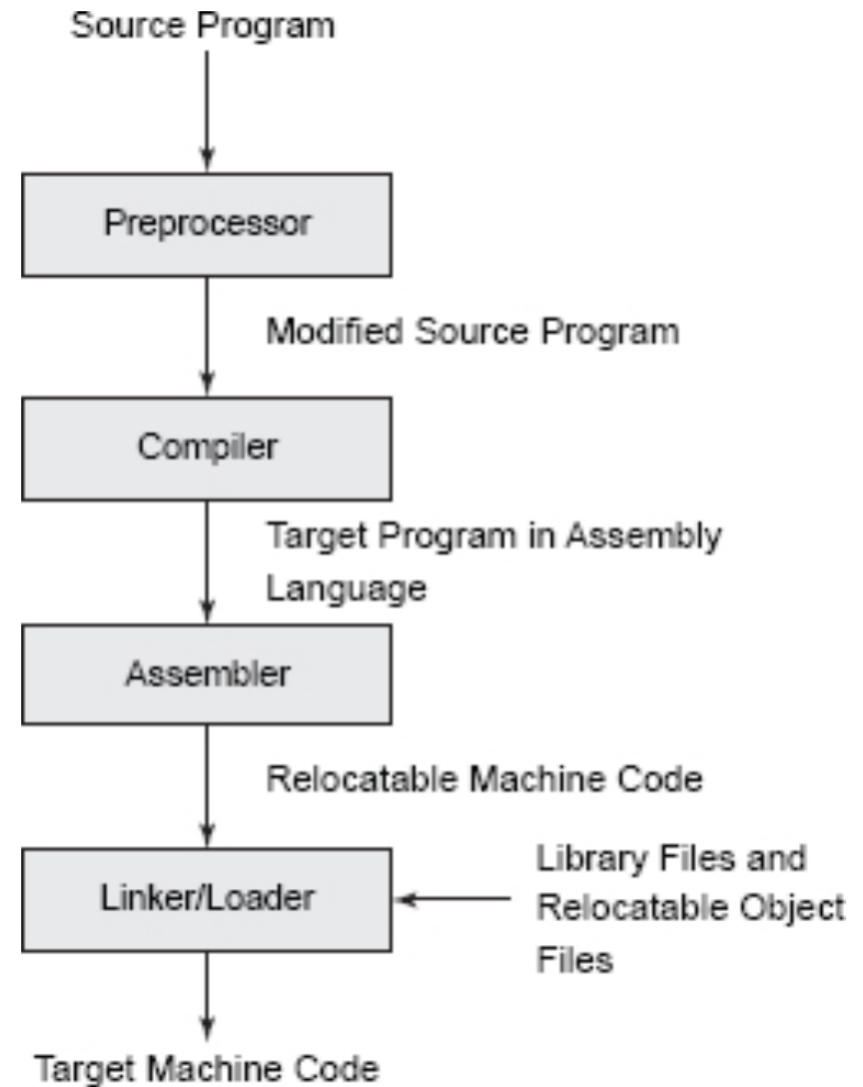
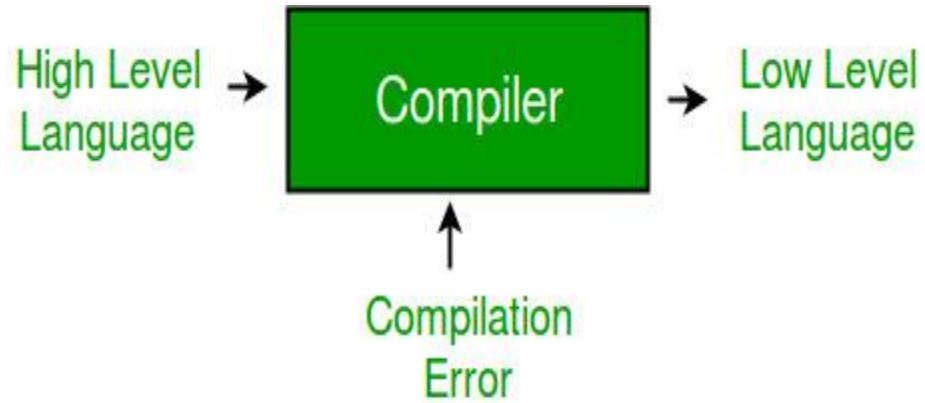
1. Identifying all the 100,000 genes in human DNA
2. Determining the sequences of the 3 billion chemical base pairs that make up human DNA
3. Storing this information in databases, and developing tools for data analysis.

Each of these steps requires sophisticated algorithms





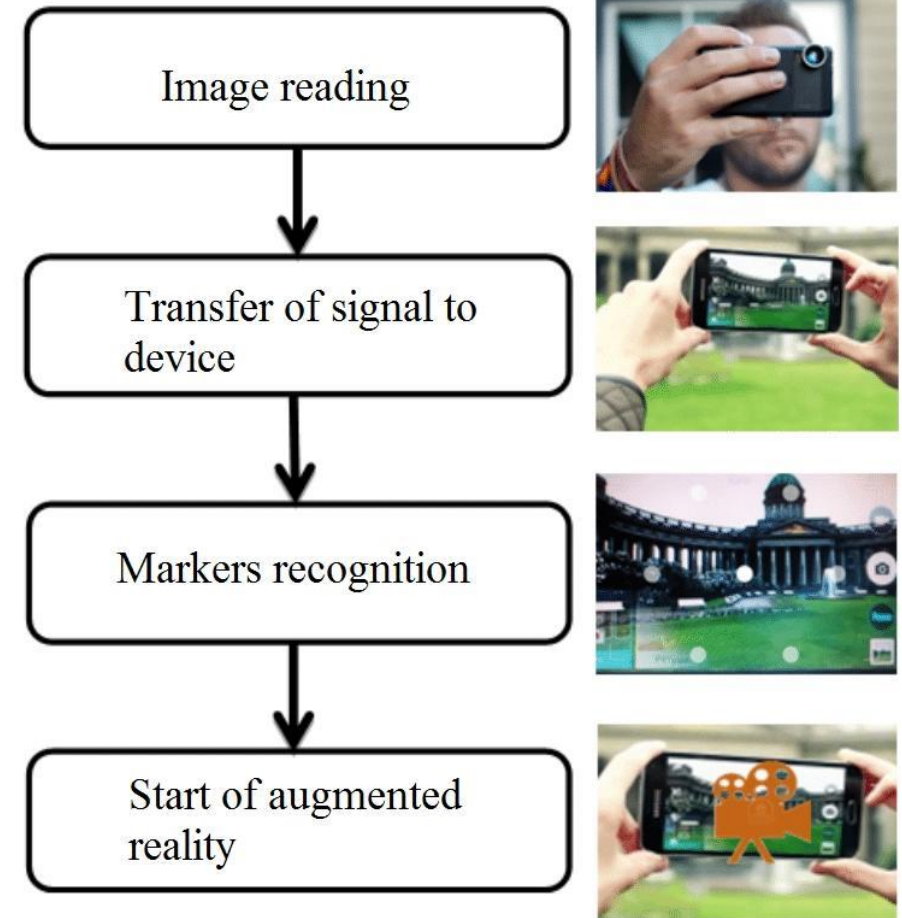
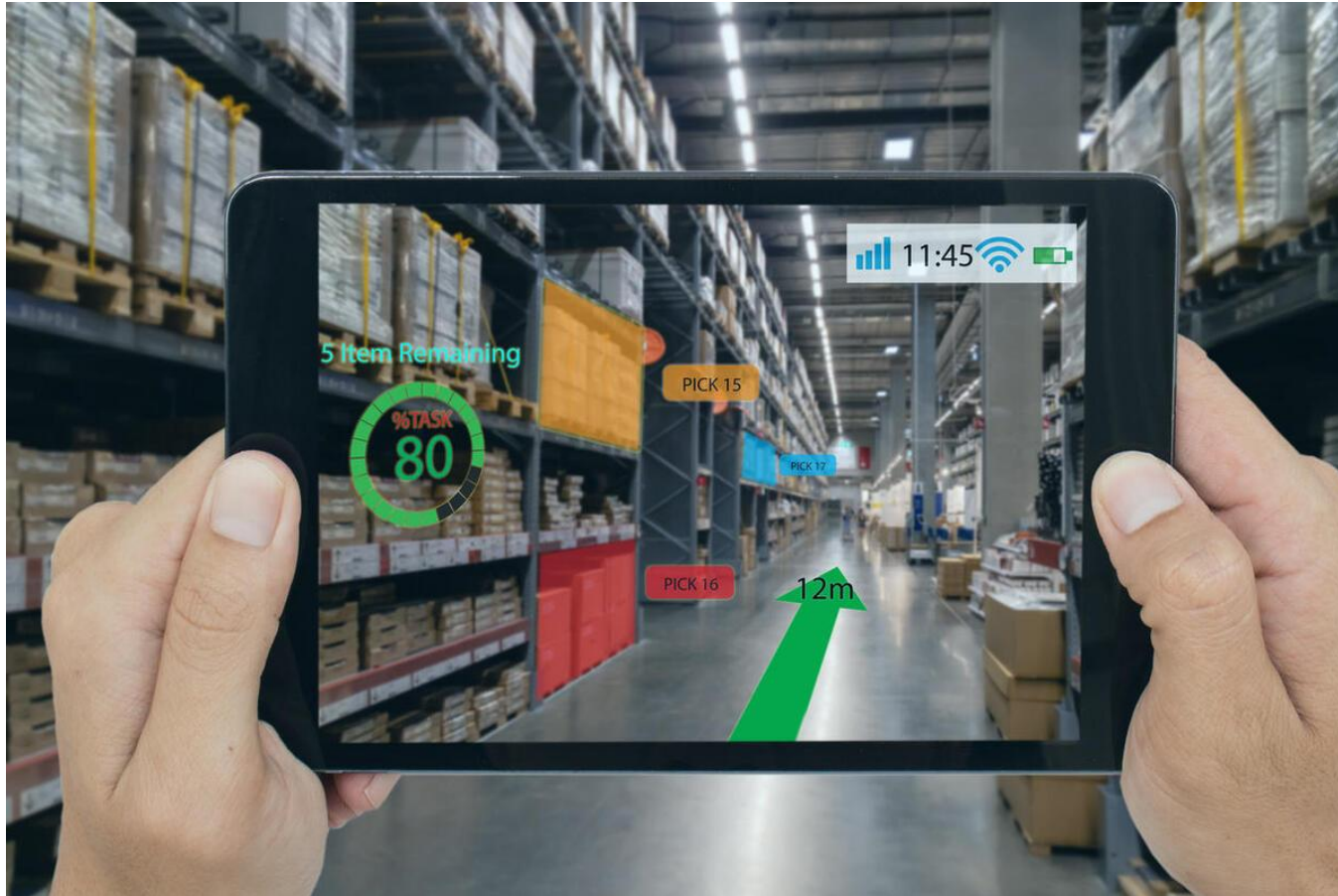
Compilers



Video games



Virtual reality



Robot Tour Optimization

- Manufacturing circuit boards, all the chips and other components must be fastened onto the substrate.
- More specifically, each chip has a set of contact points (or wires) that must be soldered to the board. To program the robot arm for this job, we must first construct an ordering of the contact points so the robot visits (and solders) the first contact point, then the second point, third, and so forth until the job is done.
- The robot arm then proceeds back to the first contact point to prepare for the next board, thus turning the tool-path into a closed tour, or cycle.

The Problem Definition

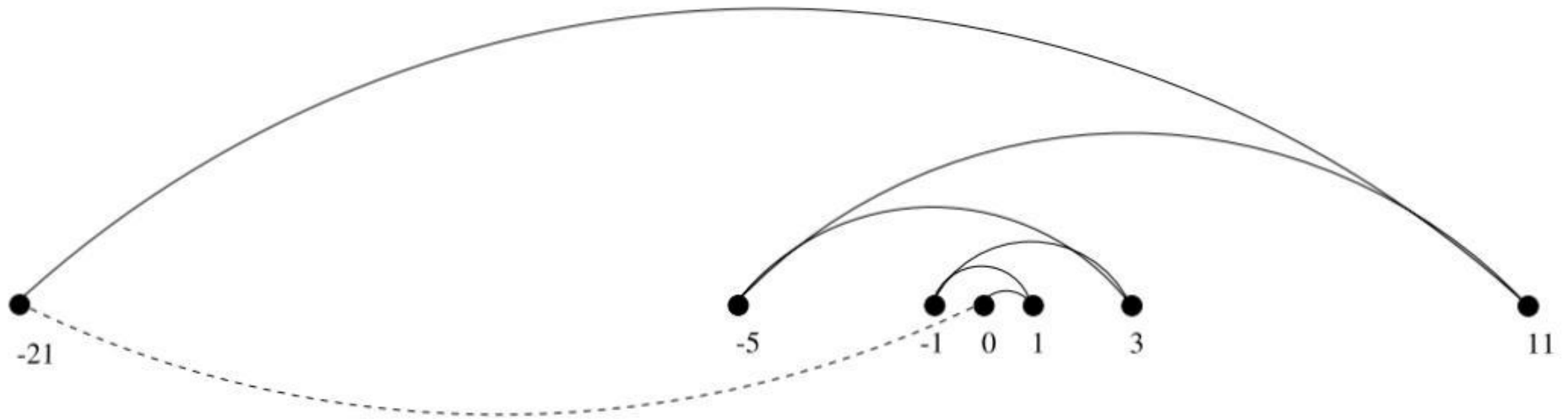
- Problem: Robot Tour Optimization
- Input: A set S of n points in the plane.
- Output: What is the shortest cycle tour that visits each point in the set S ?

Any Idea?

One Solution

- Several algorithms might come to mind to solve this problem. Perhaps the most popular idea is the nearest-neighbor heuristic.
- Starting from some point P_0 , we walk first to its nearest neighbor P_1 .
- From P_1 , we walk to its nearest unvisited neighbor, thus excluding only P_0 as a candidate.
- We now repeat this process until we run out of unvisited points, after which we return to P_0 to close off the tour.

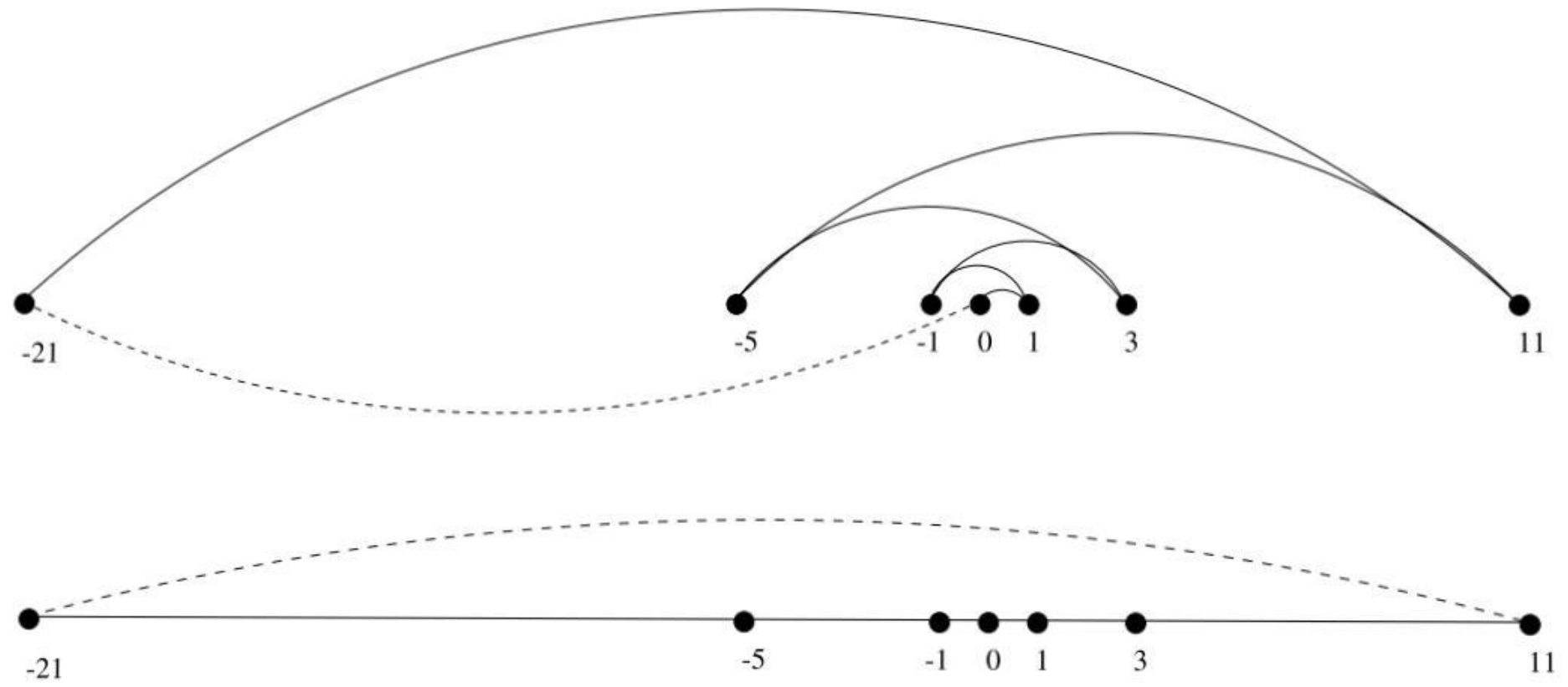
One Solution



Algorithm is not correct!

- Wrong? How can it be wrong?
- The algorithm always finds a tour, but it doesn't necessarily find the shortest possible tour.
- It doesn't necessarily even come close.
- This algorithm trades the quality of the solution for the running time.

Example



Coursework & grading (1)

- Attendance is mandatory in Zoom scheduled lectures
- Students have to watch video of the session before attending to the class
- The midterm exam will be given online via UR Courses during the regular lecture meeting time.
- The final exam is cumulative and closed book. It will be conducted online via UR Courses, and using the Proctortrack platform

Coursework & grading (2)

Time of lectures: M 1:00 - 2:15 pm (via Zoom)

Office hours: MW 2:30 - 4:00 pm (via Zoom)

Text: [Data Structures and Algorithms Analysis in C++, 4th edition, by Mark Allen Weiss](#)

Method of Evaluation :

Assignments (4 x 5%) 20%

Midterm Exam 30%

Final Exam 50%

Topics (1)

1. Review on Algorithm Analysis & Trees

- Mathematics review. Asymptotic notations and growth of functions. Overview of general, binary and search trees. AVL trees, Splay trees, B-trees.

2. Priority Queues

- Binary heap, d-heap, Min-Max heap, leftist heap, binomial heap.

3. Sorting and Order Statistics

- Overview and classification of sorting algorithms, insertion sort, merge sort, heapsort, quicksort, lower bound for sorting, bucket sort and radix sort, indirect sorting and external sorting, Order statistics.

Topics (2)

4. Graphs and Networks

- Implementation of graphs, topological sort, shortest path, network flow problems, minimum spanning tree, search techniques, introduction to NP-completeness.

5. Algorithm Design Techniques

- Combinatorial optimization, greedy algorithms, divide and conquer, dynamic programming, randomized algorithms, backtracking algorithms, branch and bound.

6. Advanced Algorithms

- Distributed Algorithms. Parallel Algorithms. Advanced algorithmic analysis: Amortized analysis, online and offline algorithms.

NP-complete Problems!

- Most of this course is about efficient algorithms. Our usual measure of efficiency is speed, i.e., how long an algorithm takes to produce its result. There are some problems, however, for which no efficient solution is known.

Parallelism

- For many years, we could count on processor clock speeds increasing at a steady rate. Physical limitations present a fundamental roadblock to ever-increasing clock speeds, however: because power density increases superlinearly with clock speed, chips run the risk of melting once their clock speeds become high enough.
- In order to perform more computations per second, therefore, chips are being designed to contain not just one but several processing “cores.” We can liken these multicore computers to several sequential computers on a single chip; in other words, they are a type of “parallel computer.”
- In order to elicit the best performance from multicore computers, we need to design algorithms with parallelism in mind.

My experience

- I used algorithms directly in medical devices including ventilators, automated external defibrillator (AED), and IoT.



Ventilator



AED



AED Box